

Communicable Disease Surveillance in Toronto: Trends and Implications for Prevention*

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Communicable disease surveillance programs can help public health agencies track changes in reported illness and plan prevention efforts around recurring seasonal patterns. This report examines Toronto's Monthly Communicable Disease Surveillance Data from 2021 to 2025, comparing monthly case totals across six disease categories and annual influenza rates. The categories show different seasonal patterns; influenza closely follows the winter peaks in vaccine-preventable disease cases and reached an annual rate of 387.0 cases per 100,000 in 2025. These findings show how surveillance data can support monitoring across diseases and help time influenza vaccination outreach before the winter increase.

1. Introduction

Communicable disease surveillance plays an important role in understanding how infectious diseases affect a population over time. By tracking reported cases over time, public health agencies can identify emerging patterns and support decision-making related to public health programming (City of Toronto 2025). In Toronto, Toronto Public Health (TPH) publishes a monthly surveillance dataset describing reported cases of a wide range of communicable diseases (Toronto Public Health 2026).

This report analyzes TPH's Monthly Communicable Disease Surveillance Data for the years 2021 through 2025, excluding data associated with COVID-19. It compares monthly reported cases across six disease categories, examines influenza's contribution to vaccine-preventable disease peaks, and considers annual influenza rates alongside national vaccination coverage estimates.

Because the timing and volume of reported cases differ by disease, annual totals alone cannot show when outreach may be most useful or which disease accounts for a category's increase.

*All code and data are available at: <https://github.com/JoshECampbell/Toronto-Communicable-Disease-Report>.

Comparing monthly patterns across categories shows that vaccine-preventable cases peak in winter, and that reported cases of influenza account for most of the volume within the category. A closer look at influenza shows that its population-adjusted annual rate also rose sharply from 2024 to 2025.

These findings show how monthly surveillance can help track seasonal patterns and flag unusual increases across diseases. The influenza pattern can also help TPH plan vaccination outreach before winter activity rises. Section 2 presents the data and observed trends while Section 3 discusses their implications for disease monitoring and vaccination outreach.

2. Data

2.1 Toronto Communicable Disease Surveillance Data

Toronto Public Health (TPH) is responsible for updating and publishing a monthly dataset that describes reported cases of communicable diseases within the City of Toronto. These datasets are made available through the City of Toronto’s Open Data Portal under the title “Monthly Communicable Disease Surveillance Data” (Toronto Public Health 2026).

As outlined by TPH, health surveillance programs help monitor public health at the population level. They can also provide evidence to inform policy decisions aimed at reducing health risks and improving population health outcomes (City of Toronto 2025). Similarly to TPH, Public Health Ontario (PHO) provides information on reported disease cases for a variety of infectious diseases, filtered by region (Public Health Ontario 2026). The PHO dataset serves as an alternative to the one used for this report and can be used to expand the analysis to investigate similar trends across multiple health units across Ontario.

Reports from 2021 to 2026 were downloaded using the statistical programming language R (R Core Team 2026), with select packages `dplyr`, `here` and `glue` (Wickham et al. 2026) (Hester and Bryan 2026) (Müller 2025). The analysis includes complete calendar years from 2021 to 2025, with revised 2025 annual counts and rates taken from the 2026 report.

`Python` was used to combine the reports into one dataset with a row for each disease and year (Python Software Foundation 2026). The rows correspond to reported case counts for a selected disease and year. Rows corresponding to COVID-19 were not included as they were out of the scope of this report. The cleaned dataset consists of 331 rows, including 79 unique diseases across 6 distinct categories. An outline of the variables as well as their descriptions can be found in Table 1 and a sample of the dataset can be found in Table 3.

Table 1: Fields found in the cleaned dataset with corresponding descriptions.

Variable	Description
Disease	Name of reported disease.
Disease Category	The reporting category assigned to the disease.
January-December	Reported case counts for the months of the calendar year.
Yearly Cases	Reported case counts for the year of surveillance for a given disease.
Yearly Rate	Yearly rate expressed as yearly cases per 100,000 population.
Year	Year of observation.

2.2 Monthly Disease-Category Patterns

The monthly series of cases at the disease-category level provides a detailed view of how reported communicable diseases evolve between January 2021 and December 2025. For each category, monthly case totals are constructed by summing the reported cases of diseases assigned to that group. The resulting series therefore represents reported cases rather than unique individuals, since an individual may appear more than once across diseases. The diseases found within each disease category are summarized in [Table 2](#).

Table 2: Diseases included in each category, 2021–2025.

Disease Category	Diseases
Diseases Transmitted by Direct Contact and Respiratory Routes	Group A Streptococcal disease, invasive (iGAS); Group B Streptococcal disease, neonatal (GBS); Legionellosis; Leprosy; Mpox*; SARS*; TB: Atypical mycobacterial infections*; Tuberculosis; Tuberculosis infection, latent (LTBI)*
Enteric Food and Waterborne Diseases	Amebiasis; Botulism; Campylobacter enteritis; Cholera; Cryptosporidiosis; Cyclosporiasis; Food poisoning; Giardiasis; Hepatitis A; Listeriosis; Paralytic shellfish poisoning; Paratyphoid fever; Salmonellosis; Shigellosis; Trichinosis; Typhoid fever; Verotoxin-producing E. coli infection (VTEC); Yersiniosis
Other Diseases	Blastomycosis; Candida auris*; Carbapenemase-producing Enterobacteriaceae (CPE); Creutzfeldt-Jakob disease (CJD)
Sexually Transmitted and Bloodborne Infections	AIDS; Chancroid; Chlamydia; Gonorrhea; HIV; Hepatitis B carriers; Hepatitis B cases; Hepatitis B unclassified reports; Hepatitis C; Ophthalmia neonatorum; Syphilis, early congenital*; Syphilis, infectious; Syphilis, late congenital*; Syphilis, late latent*; Syphilis, other; Syphilitic stillbirth*
Vaccine Preventable Diseases	Acute flaccid paralysis; Adverse vaccine reactions*; Chickenpox (Varicella); Diphtheria; Haemophilus influenzae, invasive (all types); Influenza; Measles; Meningococcal disease, invasive (IMD); Mumps; Pertussis (Whooping Cough); Pneumococcal disease, invasive (IPD); Poliomyelitis*; Rubella; Rubella, congenital syndrome (CRS); Smallpox*; Tetanus
Vectorborne and Zoonotic Diseases	Anaplasmosis*; Anthrax*; Babesiosis*; Brucellosis; Echinococcus multilocularis infection; Hantavirus*; Hemorrhagic fevers; Lassa fever*; Lyme disease; Plague*; Powassan virus*; Psittacosis/Ornithosis*; Q fever; Rabies; Tularemia; West Nile Virus (WNV)

* Not listed in every year from 2021 to 2025.

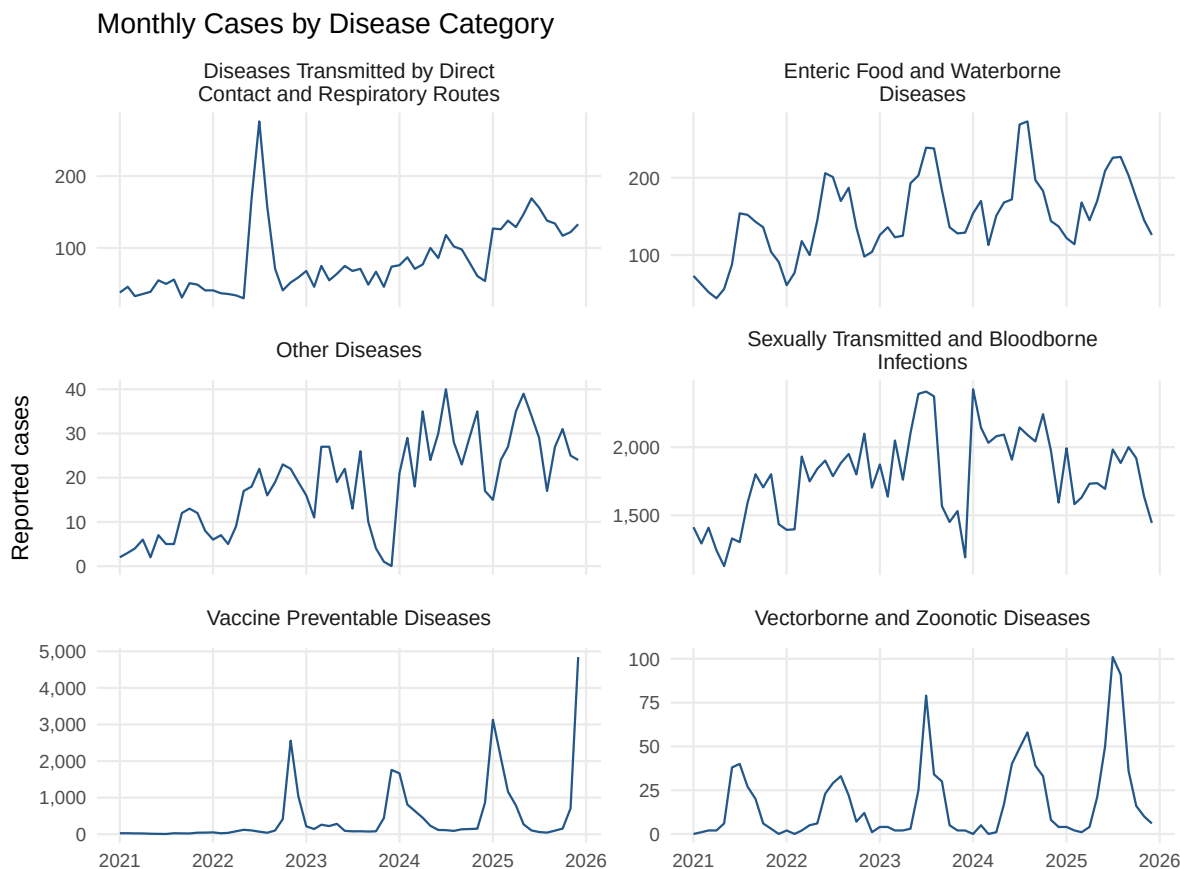


Figure 1: Monthly reported cases by category, 2021-2025. Panels use different vertical scales.

The monthly series shows substantial differences in both the volume and timing of reported cases across categories (Figure 1). Certain categories show a stronger seasonal structure. Most notably, vaccine-preventable cases are concentrated near the end and beginning of the calendar year, with distinct annual peaks occurring in December and January (December 2023, January 2024, December 2025). In contrast, enteric, food and waterborne diseases and vector-borne and zoonotic diseases rise during the summer months, with the latter peaking in July or August. The contrast shows why annual totals alone are insufficient to describe the timing of reported illness. Comparing new months with earlier seasonal patterns can flag unusual increases for closer investigation.

Comparing category totals with disease-level counts helps show which diseases account for seasonal peaks. Influenza provides an example within the vaccine-preventable category.

2.3 Influenza and Vaccine-Preventable Diseases

Between 2022 and 2025, influenza accounted for 81.4% to 94.1% of reported vaccine-preventable disease cases. Figure 2 illustrates this relationship as the monthly curve for influenza closely follows the overall category's peaks. The low influenza count in 2021 is an exception, so the pattern cannot be described as uniform across all five years. Comparing the two curves shows how much of the category's seasonal variation is associated with this single disease.

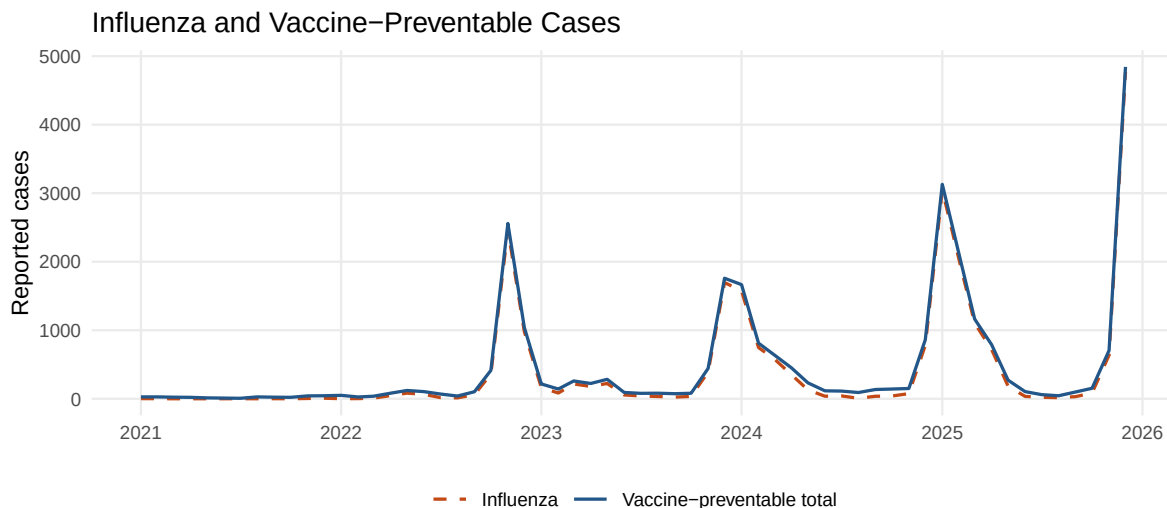


Figure 2: Monthly influenza cases and the vaccine-preventable category total, 2021-2025.

The annual influenza rate declined to 105.8 per 100,000 in 2023, then increased to 148.3 in 2024 and 387.0 in 2025, the highest value observed. The increase from 2024 to 2025 is especially noticeable, showing that much of the recent rise occurred in the final year rather than building gradually across the whole period. This jump in population-adjusted rates suggests the increased case counts during seasonal peaks cannot be completely explained by population growth.

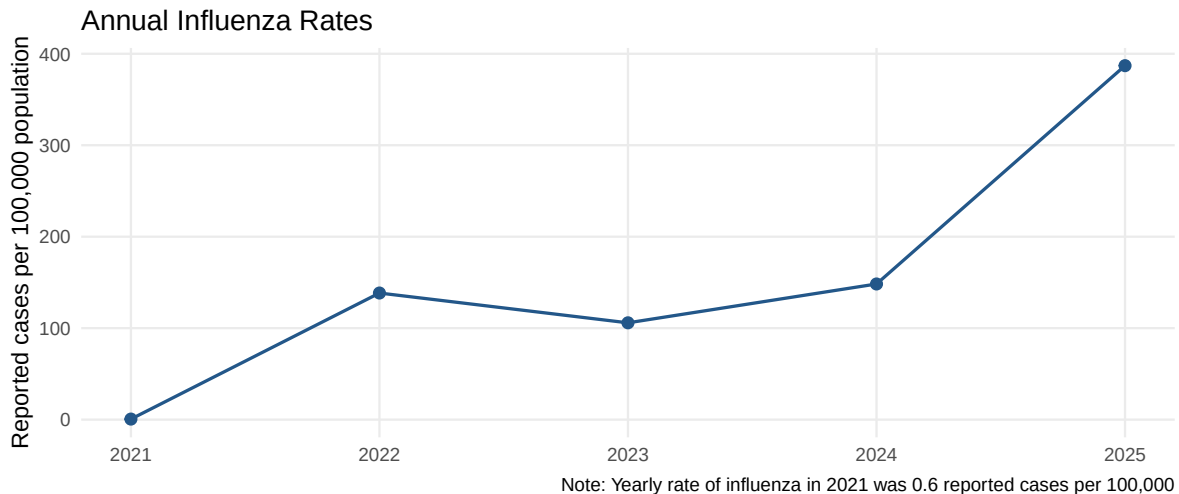


Figure 3: Annual reported influenza rates, 2021-2025.

3. Discussion

3.1 Reported Influenza Cases and Vaccination Coverage

The category comparisons show how monthly surveillance can be used across diseases. Summer increases in enteric, food and waterborne diseases and vector-borne and zoonotic diseases can be monitored against their usual seasonal timing. An unusual increase in any category can prompt closer review of individual diseases and case reports, though these patterns alone cannot establish whether an outbreak has occurred. For vaccine-preventable diseases, the recurring winter increase also provides a practical window for prevention efforts.

Influenza accounted for most reported vaccine-preventable disease cases between 2022 and 2025, and its annual rate increased sharply in 2025. The increase in population-adjusted rates indicates that population growth alone does not explain the rise in reported cases. Influenza vaccination is therefore a relevant focus for outreach before winter activity increases, although vaccination coverage is not measured in the Toronto dataset (Toronto Public Health 2026).

According to the Public Health Agency of Canada (PHAC), annual influenza vaccination is recommended for nearly everyone aged six months and older and reduces the risk of infection and flu-related complications. Vaccination also helps limit transmission to others, including those who are at greater risk of severe illness. PHAC identifies people aged 65 and older and adults aged 18-64 with chronic medical conditions as higher risk (Public Health Agency of Canada 2025). The national influenza vaccination coverage target is 80% for the two populations (Public Health Agency of Canada 2026b).

PHAC reports that vaccination among adults aged 18-64 without chronic medical conditions was under one third of the population from 2021 through 2025. In 2024-2025, coverage was 32.3% among adults aged 18-64 with chronic conditions and 63.2% among adults aged 65 and older (Public Health Agency of Canada 2026b).

For the 2025-2026 season, the survey found the main reasons for non-vaccination were the perception of good health, followed by “not getting round to it” (Public Health Agency of Canada 2026b). This may point to a gap in community outreach, particularly in communicating the importance of annual influenza vaccination to people who are not considered high-risk. In 2025, the Ministry of Health issued an executive notice announcing that publicly funded influenza vaccines were available in pharmacies across Ontario (Ontario Ministry of Health 2025). However, improving access alone may not be sufficient to increase vaccine uptake. Additional resources could therefore be directed toward reducing barriers to participation, including targeted communication campaigns and convenient vaccination clinics available in workplaces and on campuses.

Timing also matters. Influenza’s largest spikes occurred in December 2023, January 2024 and December 2025. Protective antibody levels generally develop about two weeks after vaccination, and influenza vaccination is recommended annually (Public Health Agency of Canada 2026a). These observed peaks support planning outreach early in autumn and encouraging vaccination when eligible and available, allowing time for an immune response before seasonal activity increases. Continued monthly monitoring can show whether the timing or scale of future increases differs from earlier years.

3.2 Limitations

A limitation of this analysis is its dependence on consistent reporting and testing for the diseases recorded. Additionally, national coverage estimates may not generalize to the population of Toronto and cannot show that low vaccination uptake caused the city’s increase in influenza.

Appendix

A. Sample Data

Table 3: Sample entries from the cleaned dataset.

Category		Disease	Year	Jan	Dec	Cases	Rate
Diseases Transmitted by Direct Contact and Respiratory Routes		Group A Streptococcal disease, invasive (iGAS)	2021	14	13	119	4.0
Diseases Transmitted by Direct Contact and Respiratory Routes		Group B Streptococcal disease, neonatal (GBS)	2025	0	0	5	19.1
Enteric Food and Waterborne Diseases		Salmonellosis	2025	36	30	494	15.1
Vaccine Preventable Diseases		Rubella	2025	0	0	0	0.0

Note: January and December illustrate the monthly fields. Rates are per 100,000 population.

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